North South University
Department of Electrical & Computer Engineering

LAB REPORT

Course Code : EEE/ETE 111 L

Course Title: Analog Electronics-I Lab

Course Instructor: MEZ

Experiment Number: 04

Experiment Name:

Clipper and Clamper circuits

Date of Experiment:

Date of Submission:

Section: 03

Group Number: 7

Submitted By	Score

	Student Name	ID
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3.		
4.		
5.		

Experiment : 04

Experiment Name: Clippen and Clampen circuit.

Objective: Study of Clippen and Clampen circuit.

Theory:

Clippen: Clippens are networks that employ diodes to 'clip' away a portion of an input signal without distorting the remaining part of the applied waveform. Types of Clippen Diode circuit - positive diode Clipping circuit, Negative diode Clipping circuit, Clipping of both cycle, Biased Diode clipping circuit.

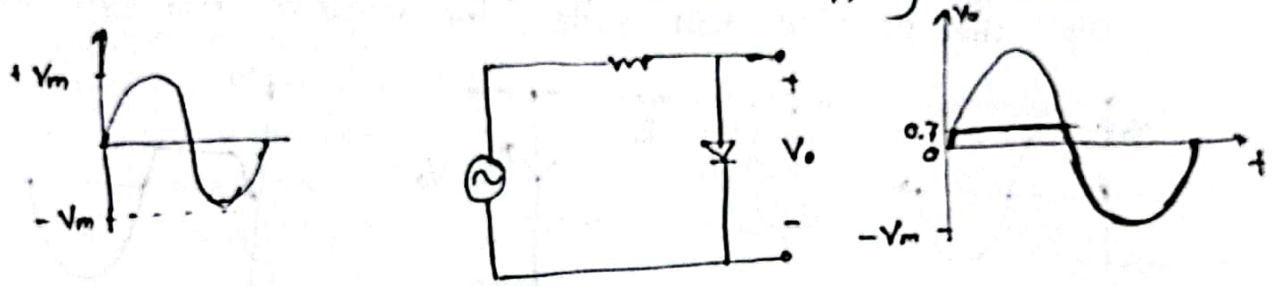


Fig. 1

In this diode dipping circuit the diode is forward biased the diodes begins to conduct and holds the voltage across itself constant at 0.7v until the sinusoidal waveform falls below this value. During the positive half cycle.

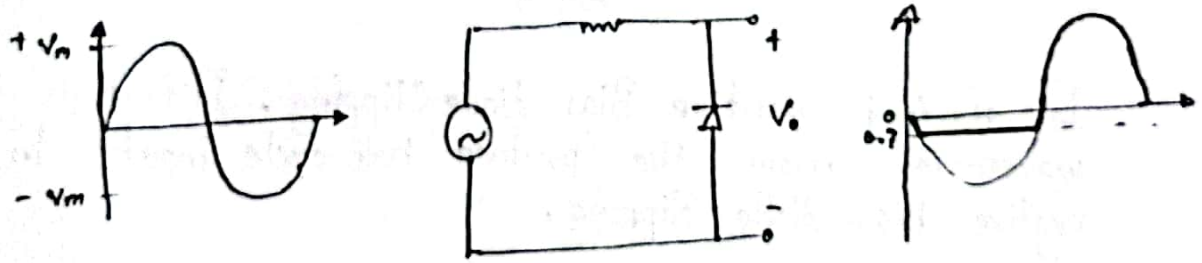


Fig. 2

The diode is forward biased during the negative half cycle of the sinusoidal waveform and limits or clips it to -0.7 v while allowing the positive half cycle to pass unaltered when reverse biased.

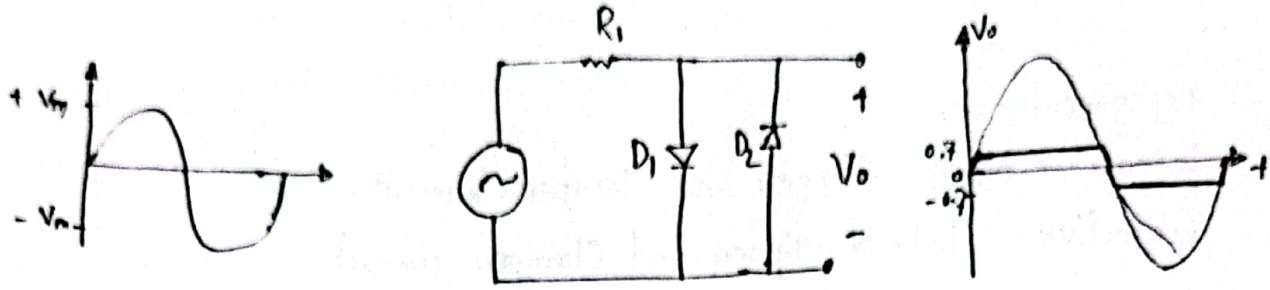


Fig 3

Connected two diode in inverse parallel as shown, then both the positive and negative half cycle would be clipped as diode D_1 clips the positive half cycle of the sinusoidal input clip the positive waveform wil while diode D_2 clips the negative half cycle. Then diode clipping circuits can be used to clip the positive half cycle, the negative half cycle or both.

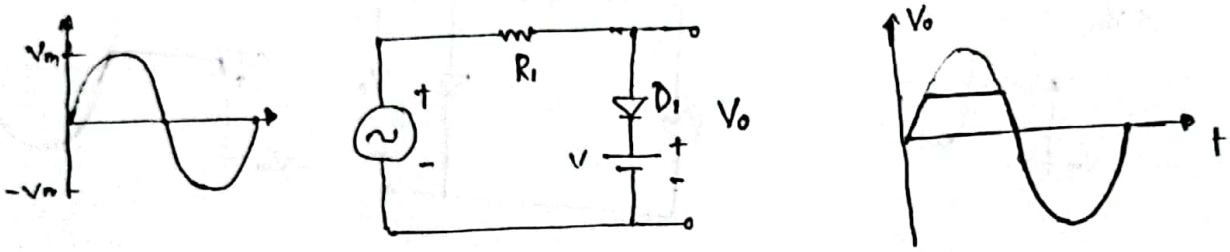


Fig : 4

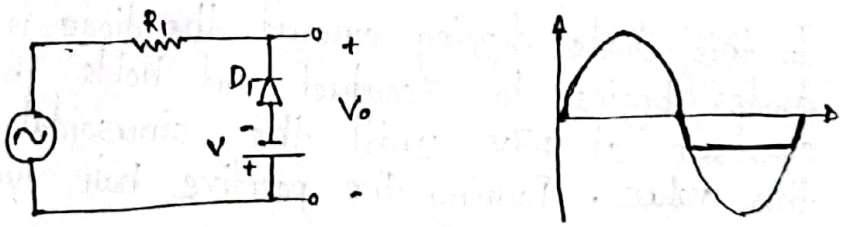


Fig 5

In fig 4 is positive Bias diode Clipping. It's clips the positive waveforms from the positive half cycle input. In fig 5 is negtive bias diode clipping.

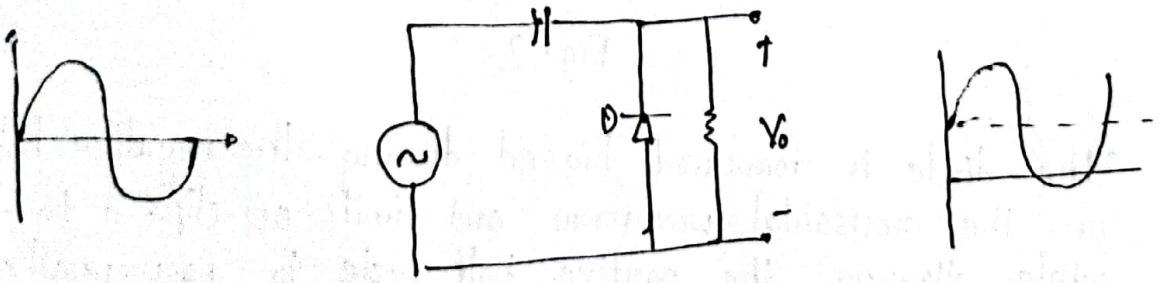


Fig 6

Clamper: A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different dc level with out the appearance of the applied signal A clamper does is adding a DC component to the signal. In fig 6 the input signal is a sinewave, the clamper pushes the signal upward, so that negative peaks fall on the ov level. As can see the shape of the original signal is preserved all that happen is a vertical shift of the signal.

Equipment list:

Se. No.	Component Details	Specification	Quantity
1	p-n junction diode	IN4007	1
2	Resistor	100 K-Ω	1
3	capacitor	0.1MF	1
4	signal Generator		
5	Trainer Board		
6	DC Power supply		
7	oscilloscope		
8	DMM		
9	wa chords and wire		as required

Circuit Diagram:

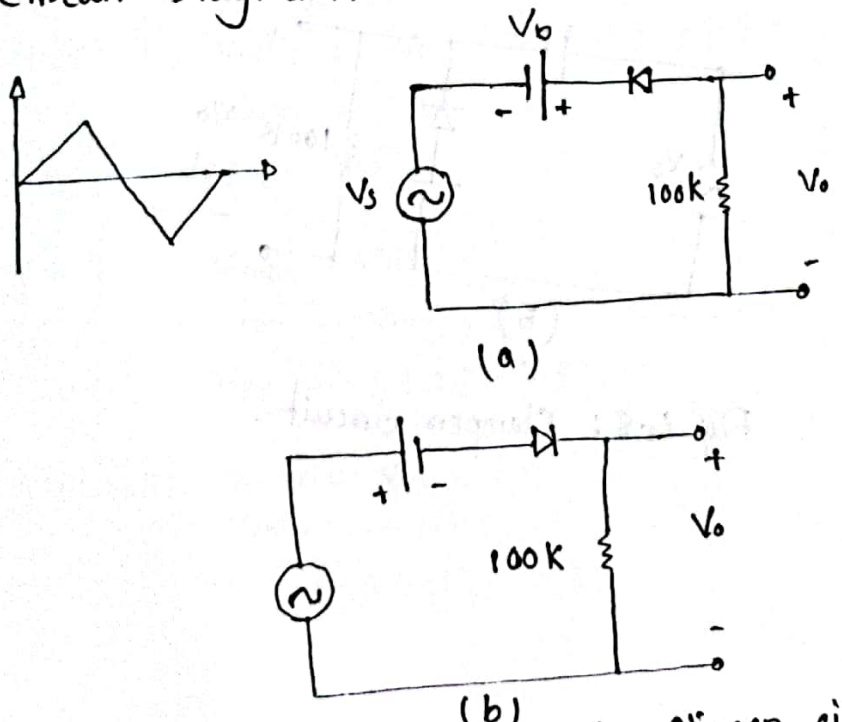
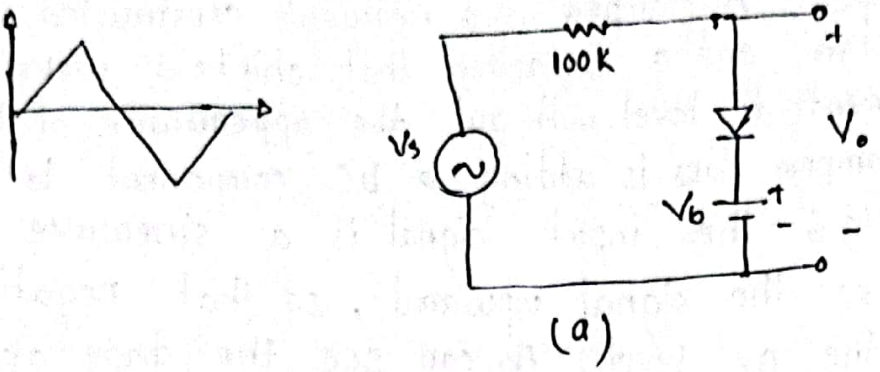


Fig 4.6: Series Clipper circuit.



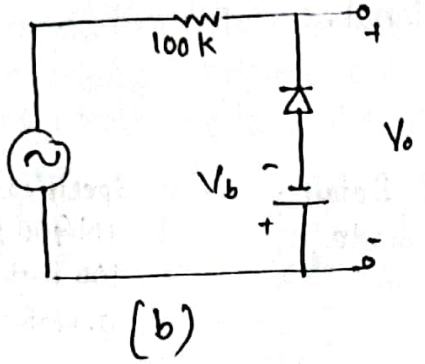
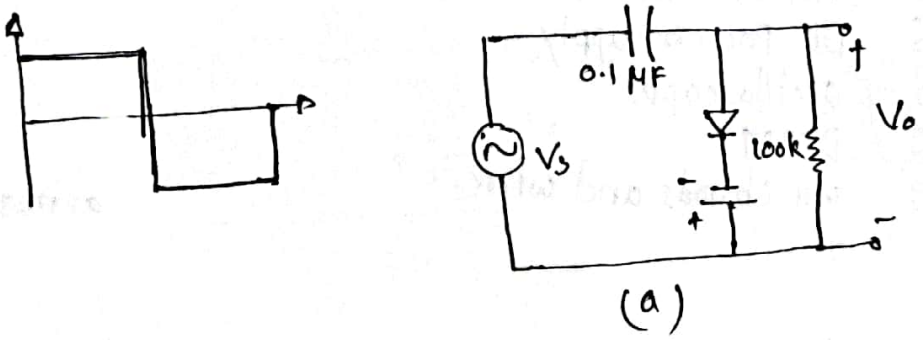


Fig 4.7: Parallel Clippers Circuit.



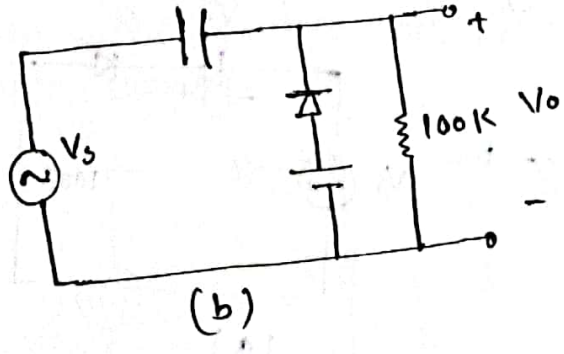


Fig 4.8: Clampers circuit.

Data and Results:

Figure Number	V_b (v)	Practical Value of V_{pp}	Theoretical Value of V_{pp}	% Error
4.6 (a)	2.5	2.08	2.5	16.8 %
4.6(b)	2.5	2.4	2.5	4 %
4.7(a)	2.5	7.2	7.5	4%
4.7(b)	2.5	7.52	7.5	0.26%.

$$\% \text{ error} = \frac{|\text{Theoretical} - \text{Practical}|}{\text{Theoretical}} \times 100$$

Theoretical value were found using multisim.

Calculation:

Fig 4.6(a), $V_{max} = 0$

$V_{min} = -2.5 - (V_m - V)$

$= -(5 - 2.5)$

$= -2.5$

$V_{pp} = 2.5 + 0 = 2.5$

Fig 4.6 (b), $V_{max} = V_m - V = 5 - 2.5 = 2.5$ v

$V_{min} = 0$

$V_{pp} = 2.5$

Fig 4.7(a) $V_{max} = 2.5$ v

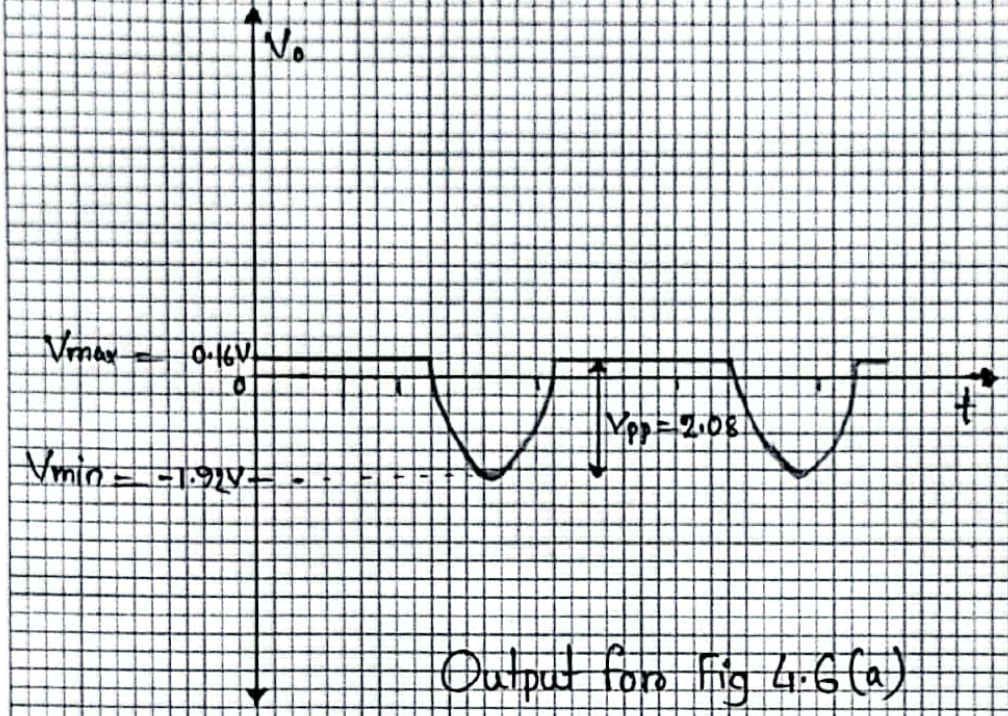
$V_{min} = -V_m = -5$ v

$V_{pp} = 5 + 2.5 = 7.5$

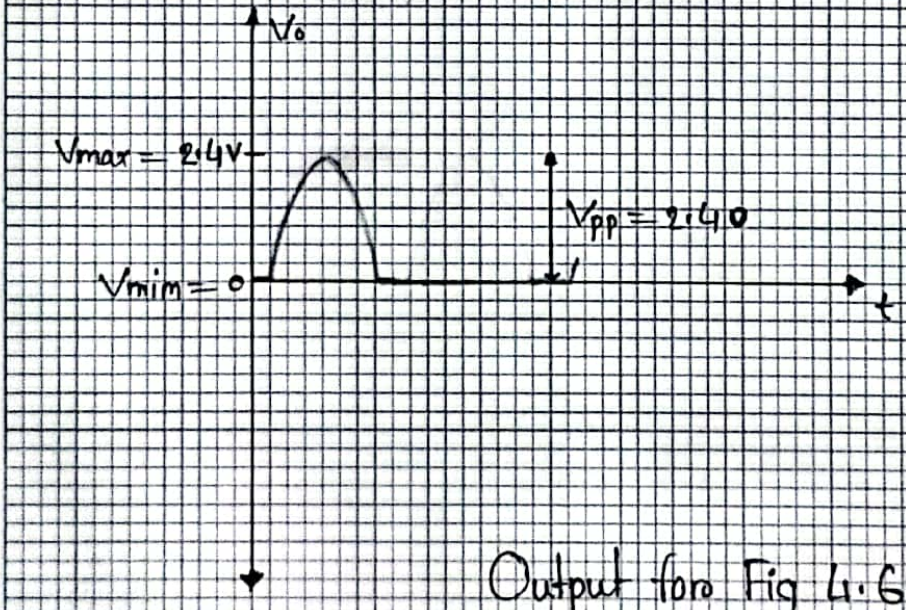
Fig 4.7(b) $V_{max} = V_m = 5$ v

$V_{min} = -2.5$ v

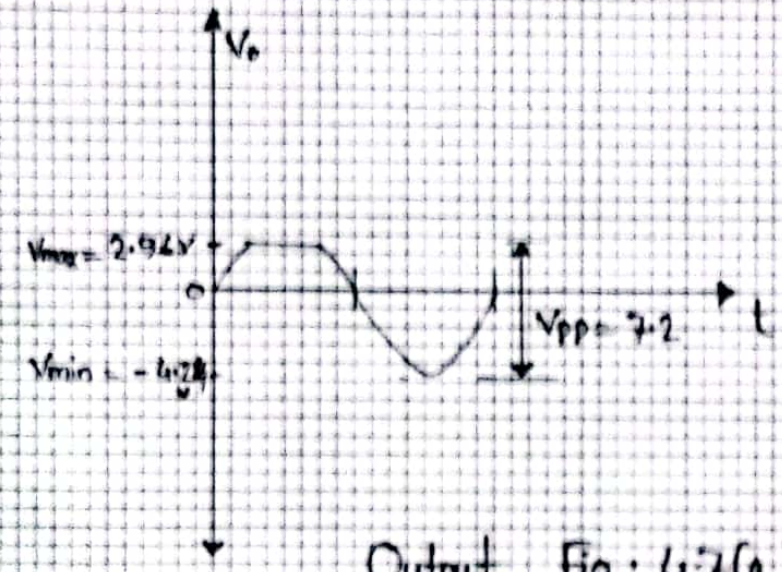
$V_{pp} = 5 + 2.5 = 7.5$



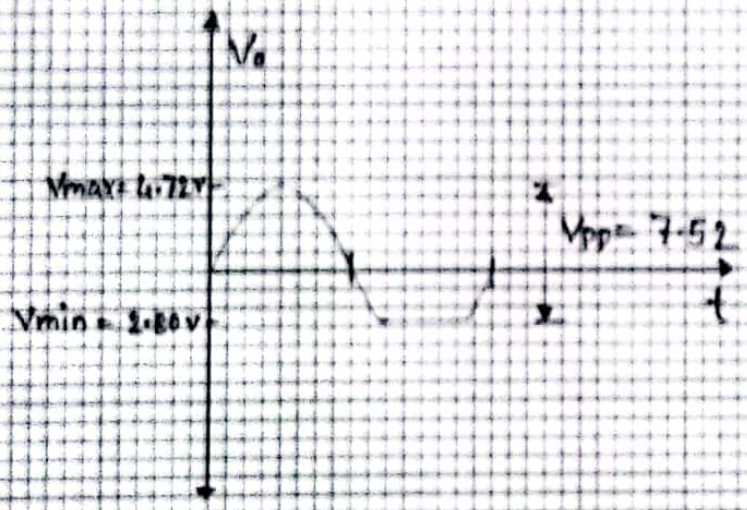
Output for Fig 4.6(a)



Output for Fig 4.6(b)

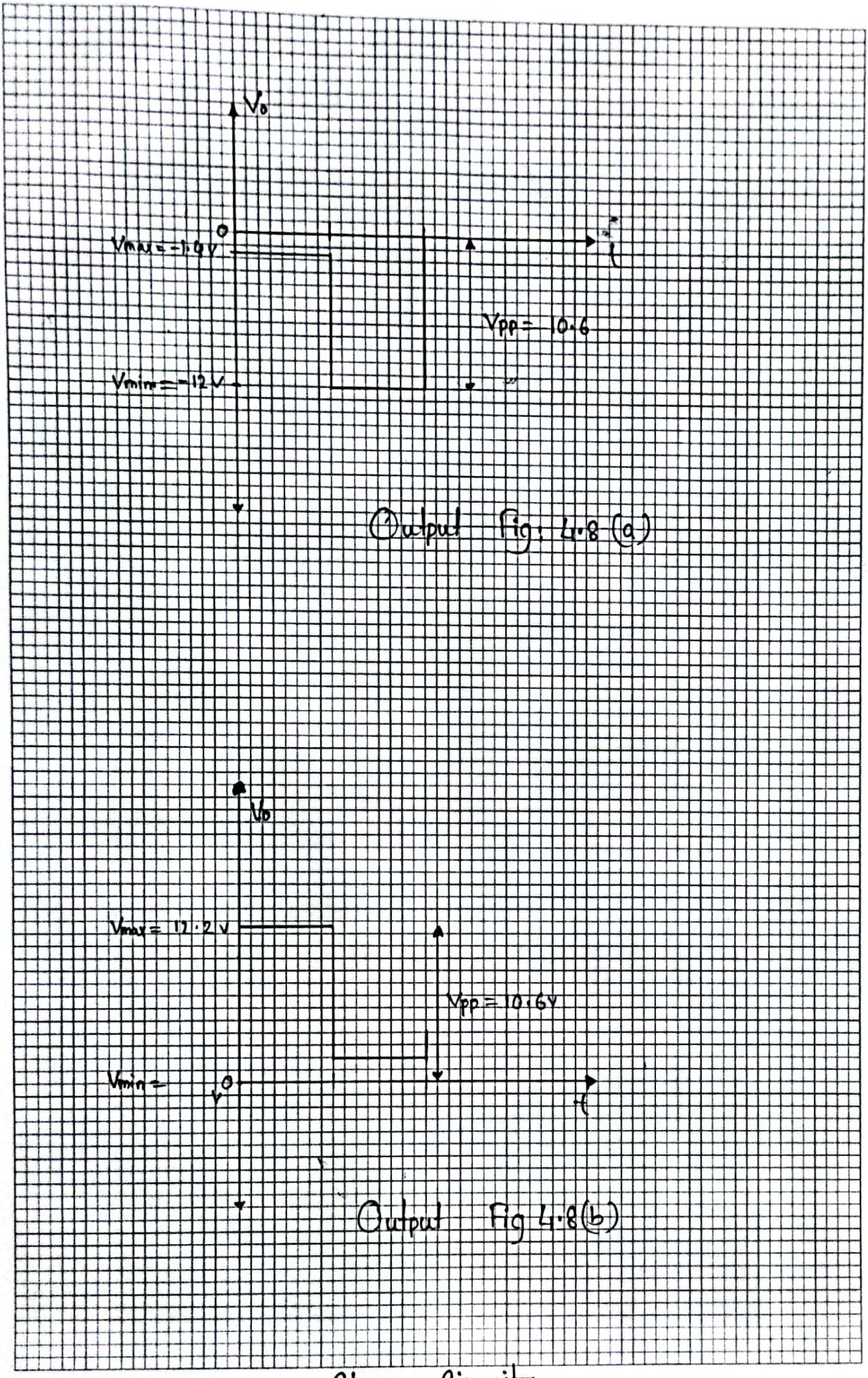


Output Fig : 4.7(a)



Output Fig 4.7(b)

Panalle Clippers Circuit

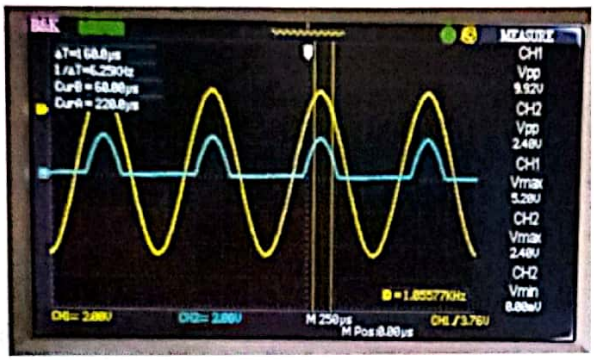


Output Fig: 4.8 (a)

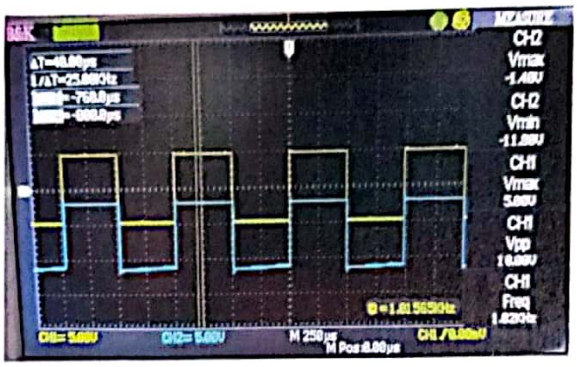
Output Fig 4.8(b)

Clampen Circuit

Output wave shape of Fig4.6 (b) when Vb=2.5 v,



Output wave shape of Fig4.8 (a) when Vb=2.5 v,



Output wave shape of Fig4.8 (b) when Vb=2.5 v,



Output wave shape of Fig4.7 (a) when Vb=2.5 v,

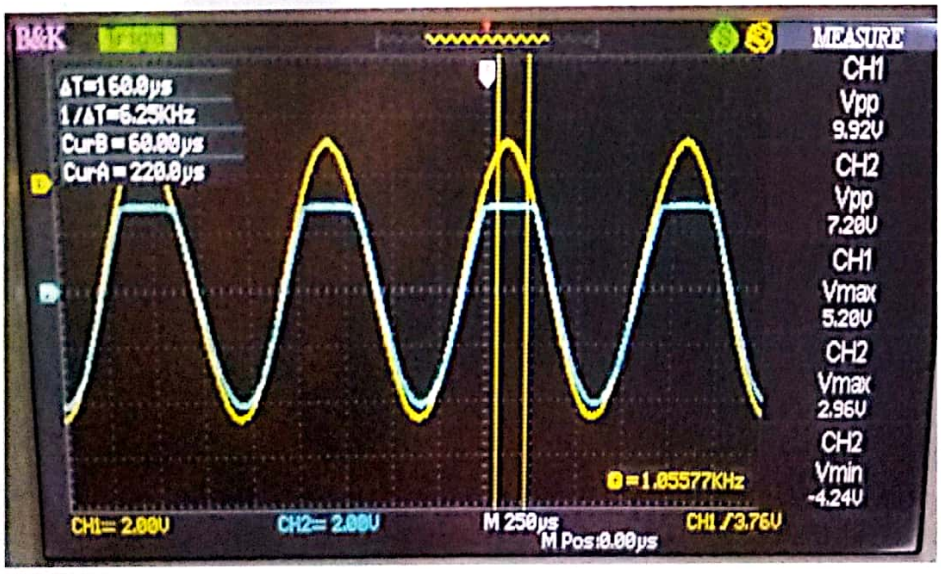


Figure : 4.3

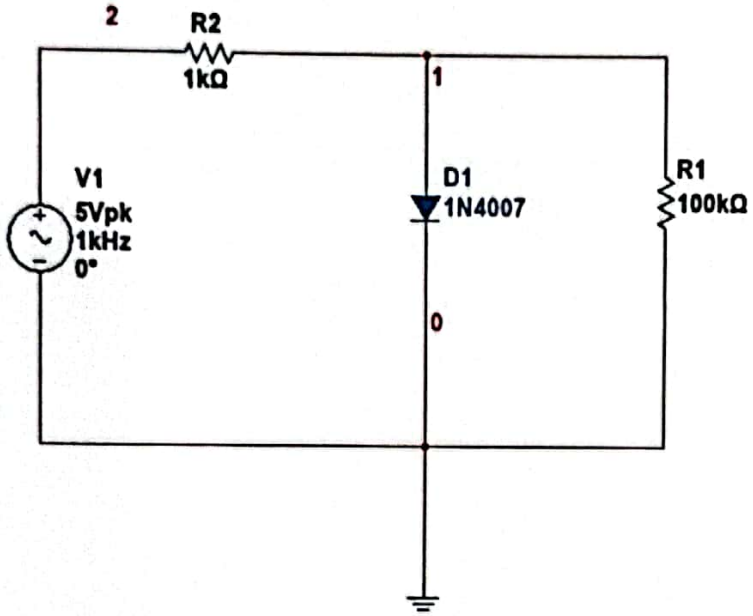
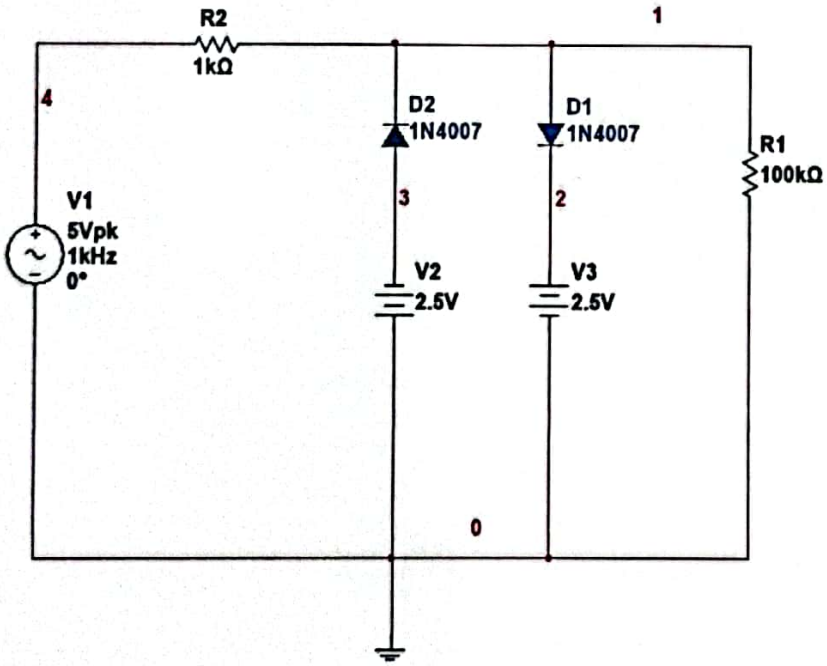
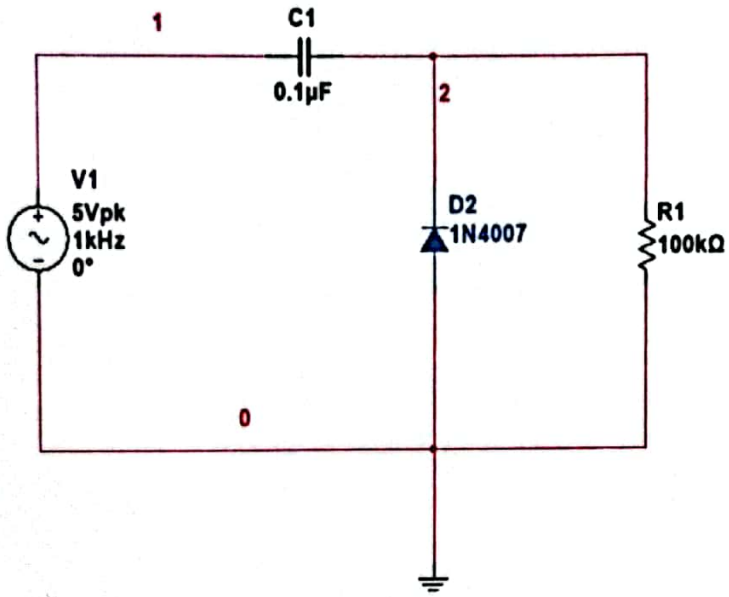


Figure : 4.4



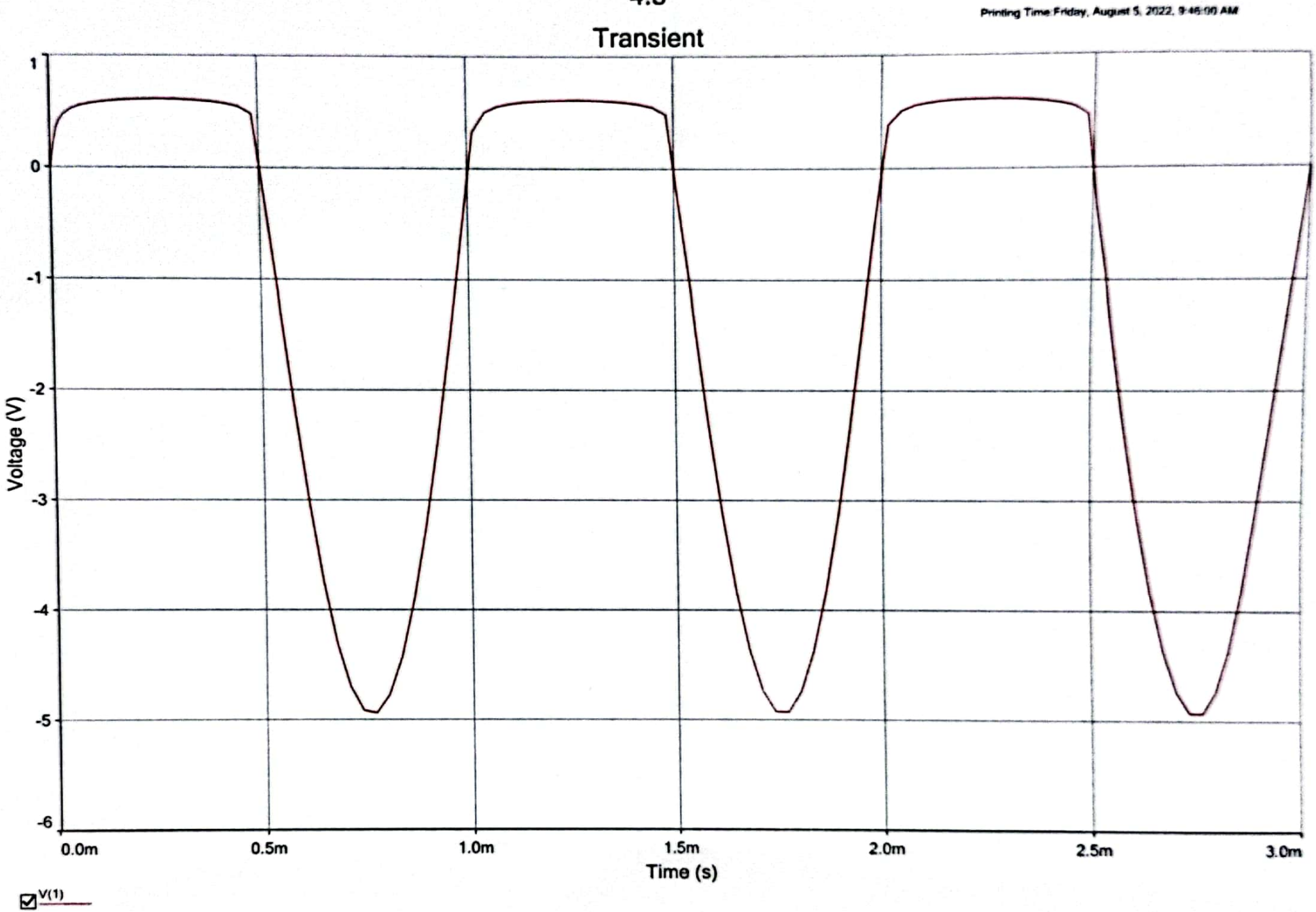
Simulation done by MOHAMMED ABDULLAH AYAZ

Figure 4.5



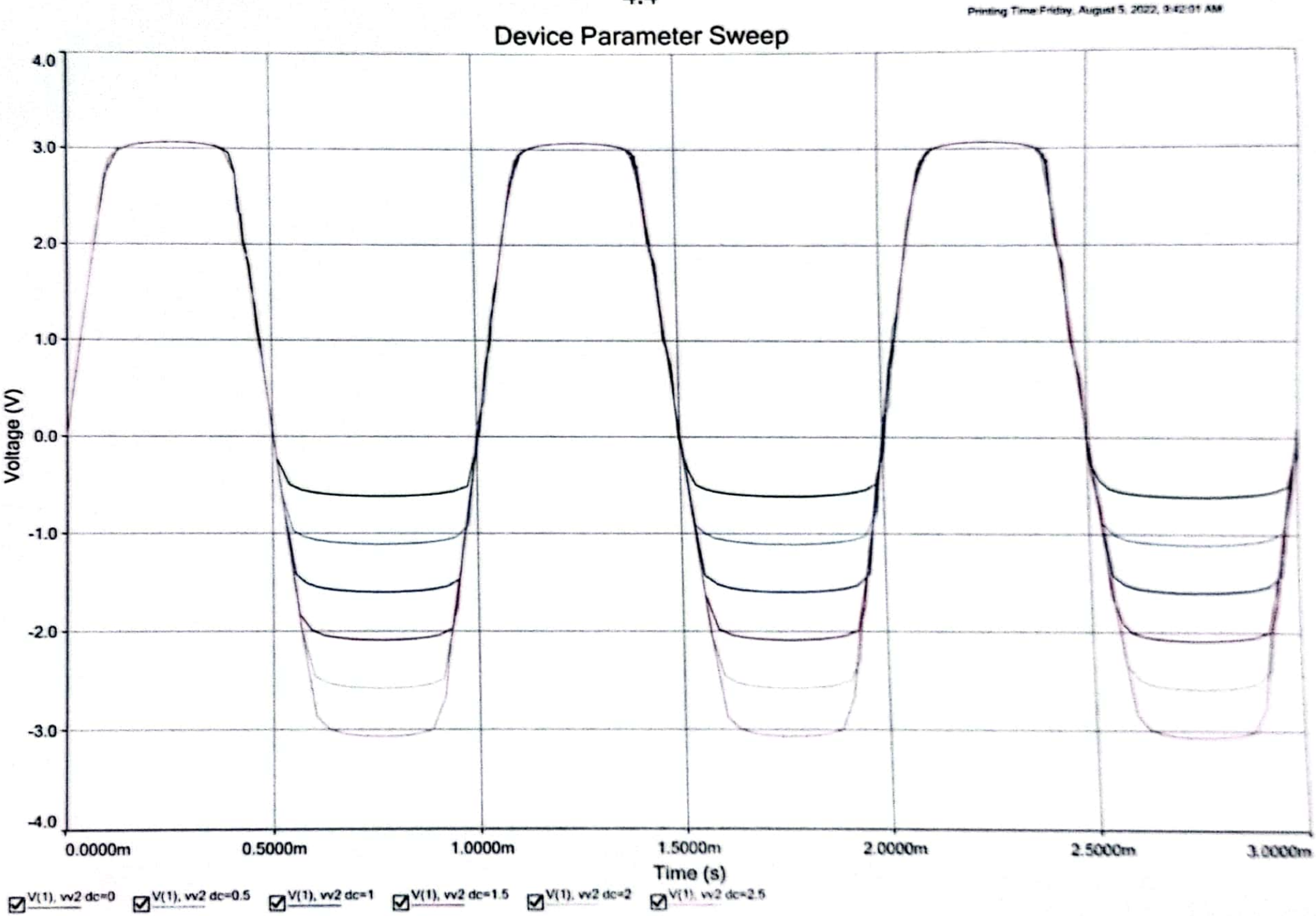
4.3

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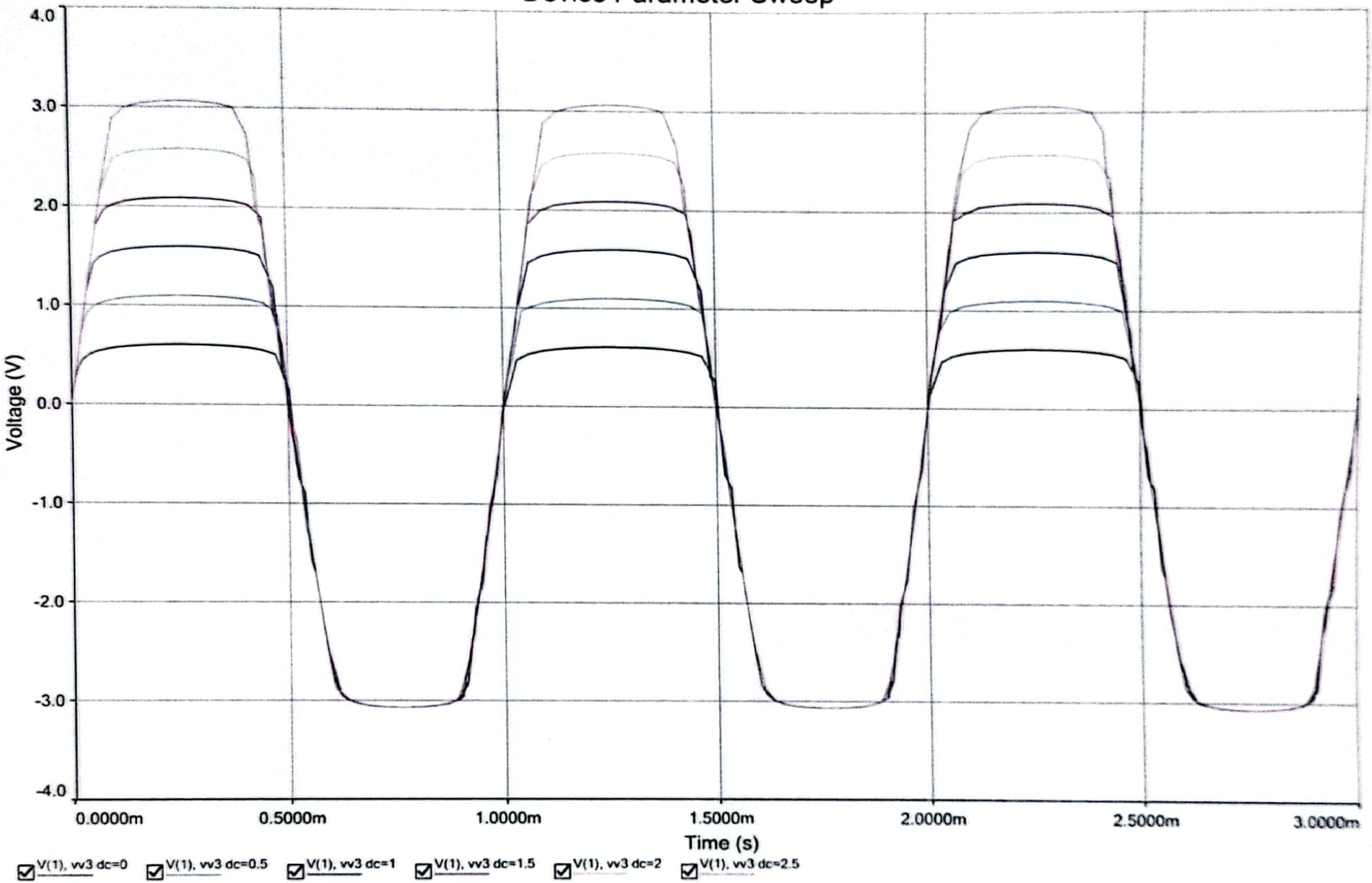
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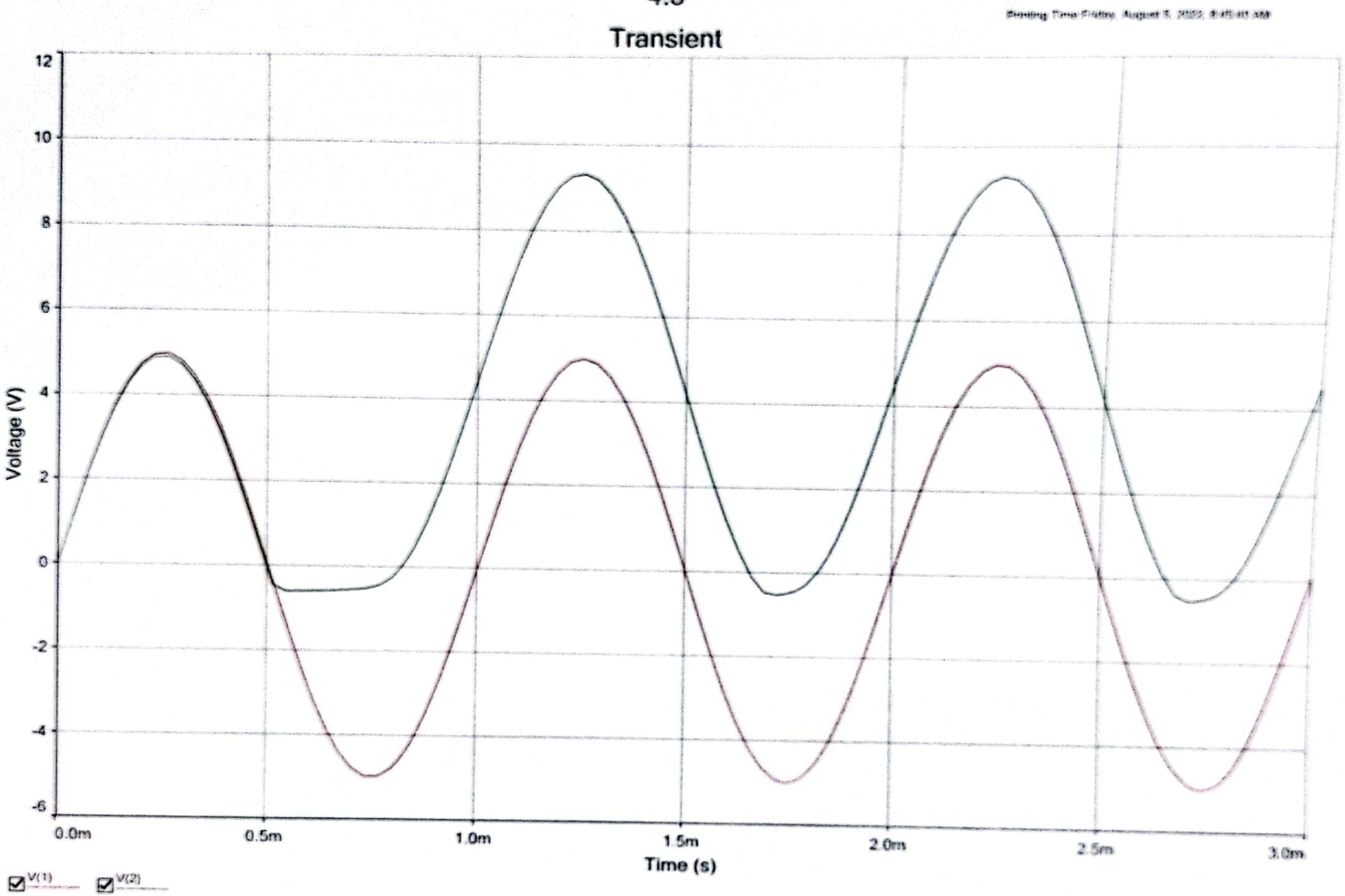


4.4

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4.5


Transient
Plotting Time: Friday, August 5, 2022, 8:45:01 AM
12
10
8
6
4
2
0
-2
-4
-6
Voltage (V)
0.0m
0.5m
1.0m
1.5m
2.0m
2.5m
3.0m
Time (s)
V(1)
V(2)

Result Analysis and Discussion:

In this experiment we were able to analyze the output voltage graphs of both clipper and clamper circuit. It's behave the V_b voltage increasing and decreasing And observ. Output voltage graph changes.

In the theoretical value we observ that in the series clipper circuit [Fig: 4.6(a)] by the $V_b = 2.5$ volts. V_o is $V_{max} = 0\,V$, $V_{min} = -2.5\,V$. So that V peak to peak voltage is 2.5 volts. When get into practically the $V_{pp} = 2.08$. The percentage of error is 16.8%. The % Error for the Fig 4.6(a) and the 4.7(a) are same. The lowest % Error of Fig 4.7(b) is 0.26%. And the heighest value of % Error is 16.8%.

In fig 4.6(a) we see that when we decrease the V_b to 2.5 V_{pp} voltage increases. As same as that in Fig 4.6(b). In fig 4.6(a) when we increas V_b to 2.5 V to 5. We see that V_{pp} is decressing. As same as that in fig 4.6(b). In fig 4.7(a) and 4.7(b) when decressing V_b voltage we see that the V_{pp} voltage is decreasing and when we increasing V_b Voltage to 2.5 to 5 the V_{pp} Voltage as well as same result.

In this experiment we get when we get meet few problems. so avoid such problem on erros everyone must take precaution and change the components to their respective.

Question And Answer:

1. —

2. In fig 4.6(a&b) ~~ob~~ we decrease the value of V_b and observe that output voltage V peak to peak is increasing. cause 4.6(a&b) circuit V_b is in reverse bias of the diode, so it will increasing the V_{pp}.
In fig 4.7 (a&b) we decrease the value of V_b and observe that the V_{pp} is decreasing with V_b voltage. For the fig 4.8 (a&b) clamper circuit, when we decreasing the value of V_b we observe that V_{pp} value as same as the previous result. It's behind ~~by~~ the capacitor we gets the same results.

3. For the series clipper circuit when we increasing the value of V_b, V_{pp} value decreasing. And the parallel clipper circuits when we increasing the value of V_b we see that as per as the same result there are not too much difference. In fig 4.8 clamper circuit when we increasing the value of V_b, in fig 4.8(a) Output graph move down, Output voltage decreases. And fig 4.8(b) Output graph move up ~~and~~, output voltage increasing but we see that the V_{pp} voltage is same as the previous result.

NORTH SOUTH UNIVERSITY

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Data Collection: *Signature of instructor:*

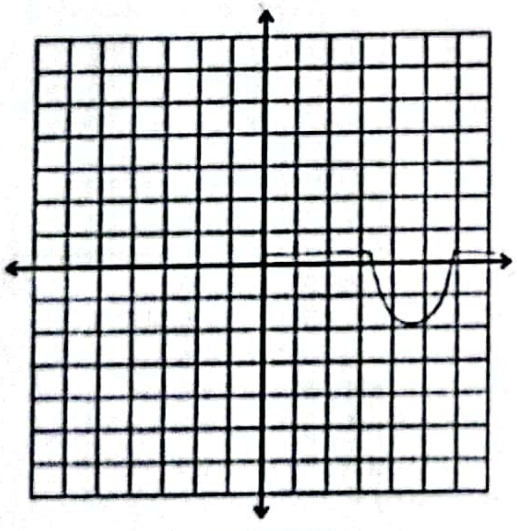
Experiment: 4,
Performed by Group# 7

Theoretical value: R = 100 kΩ

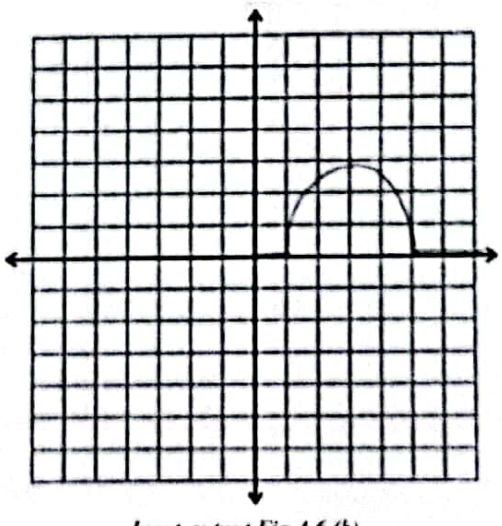
Measured value: R = 103.3kΩ

Vs = 9.92 V(p-p).

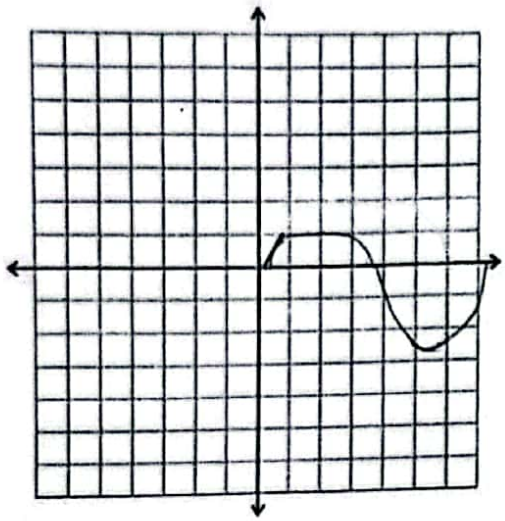
V_b (V)	Vo (p-p)					
	Fig 3.6		Fig 3.7		Fig 3.8	
	(a)	(b)	(a)	(b)	(a)	(b)
0	4.84	4.8	4.72 v	5.04v	10.6	10.6
1	3.52	3.84	5.76 v	6.00 v	10.6	10.6
2	2.64	2.88	6.72 v	7.04 v	10.6	10.6
2.5	Vmax = 160 mV Vmin = -1.92 V	Vmax = 2.64 V Vmin = 0 V	Vmax = 2.96 Vmin = -6.72 V	Vmax = 6.27 Vmin = -2.80	V_{max} = 3.4 V_{min} = -12	V_{max} = 12.2 V_{min} = 1.6 V
3	1.60	1.92	7.60 v	8.08v	10.6	10.6
4	560 mV	960 mV	8.48 v	8.88 v	10.6	10.6
5	160 mV	160 mV	8.88 v	8.88 v	10.6	10.6



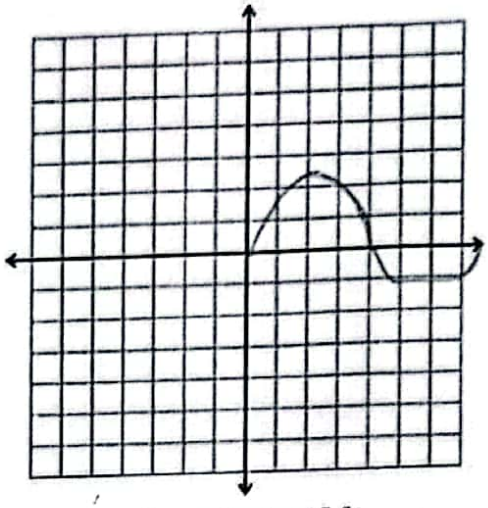
Input-output Fig 4.6 (a)



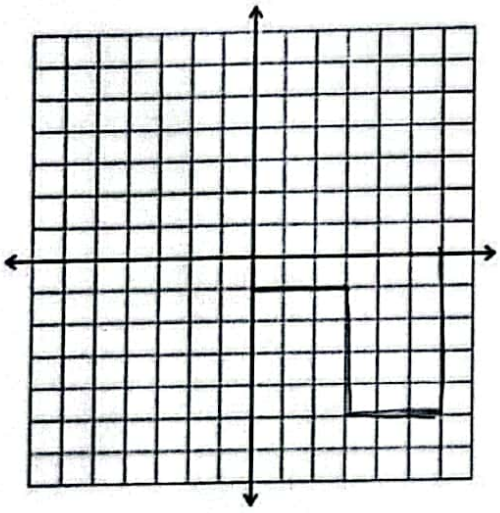
Input-output Fig 4.6 (b)



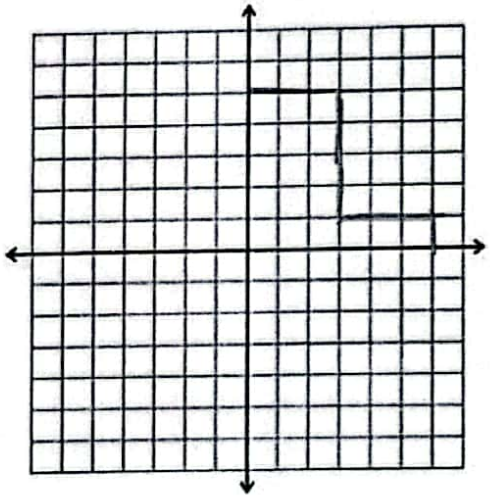
Input-output Fig 4.7 (a)



Input-output Fig 4.7 (b)



Input-output Fig 4.8 (a)



Input-output Fig 4.8 (b)

Report:

1. Using values from your data table, for all the circuit diagrams plot the input-output waveforms observed on the oscilloscope for V_b=2.5V.
2. For Fig 4.6(a &b), Fig4.7 (a & b) and Fig 4.8 (a & b) what change did you observe in the output voltage, In procedure-4? *Explain the reason behind such a change.*
3. For Fig 4.6(a &b), Fig4.7 (a & b) and Fig 4.8 (a & b) what change did you observe in the output voltage, In procedure-5? *Explain the reason behind such a change.*